

Faculty of Dental Medicine

Polymer Denture base materials & Soft liners



by

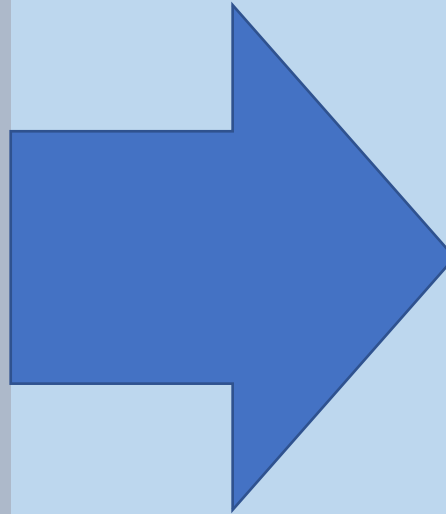
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Introduction

Polymerization chemically links small organic compounds called monomers into long chains of repeating units (polymers).

The reactions of polymerization are: **Addition** or **Condensation** according to the reaction mechanism.

the physical and chemical properties of polymers are always different from **monomers**.



For example:

methylmethacrylate, a monomer used to make denture base, is a liquid which boils at range of 100.3 ° C.

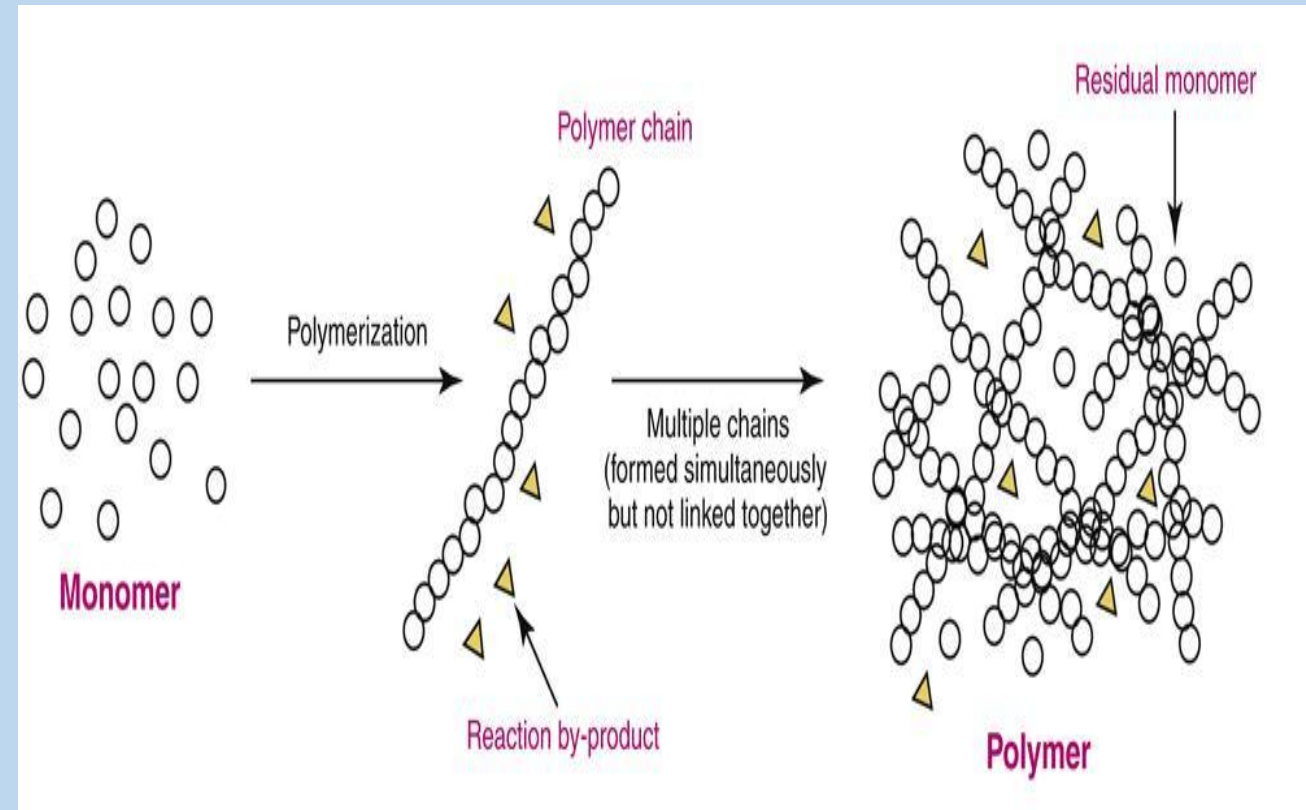
However, *poly(methyl methacrylate)* (**PMMA**)

polymers (formed from methyl methacrylate monomer polymerization) never boil but are rigid and decompose before they melt.

-Also, not all monomer reacts, leaving **residual monomer** trapped in the polymer.

Residual monomer and reaction by-products may have:

- 1- **negative clinical effect** on oral tissues (inflammation), and systemic effects and
- 2- changing the **physical properties** of the polymer itself.



<https://pocketdentistry.com/13-polymers-in-prosthodontics/>

- **Residual monomer**

- any unreacted residual monomer in the polymer may have harmful clinical effects.
- For example, residual monomer may cause allergic reactions or chemical burns

- **Degree of conversion**

- The extent to which all monomer is converted to polymer is called the degree of conversion.
- Polymers with high degrees of conversion have low levels of residual monomer; a high degree of conversion is generally desirable.

Poly Methyl Methacrylate, Acrylic Polymers (PMMA) as a Denture Base

Advantages:

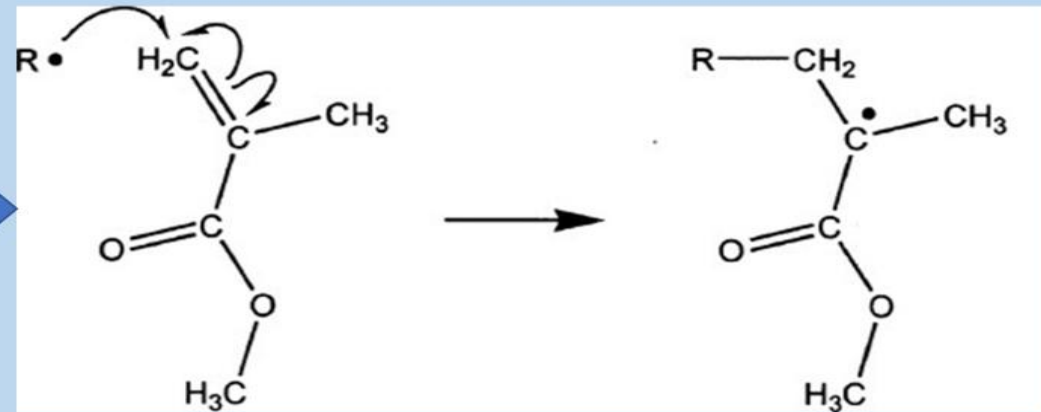
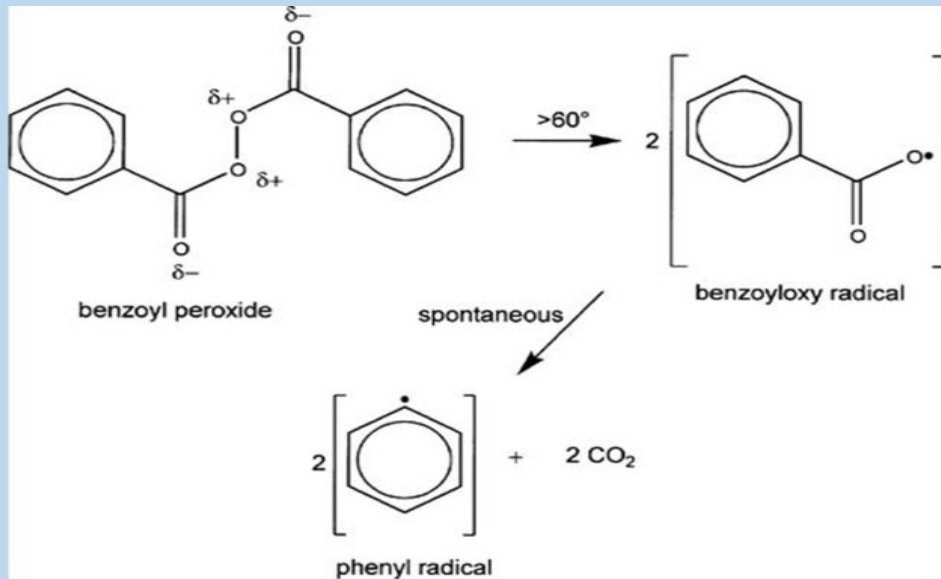
Ease of processing, acceptable mechanical properties, aesthetics and low price, so PMMA has replaced previously used denture base materials

Chemistry and polymerization:

Type of polymerization		synthetic polymer prepared by the free radical addition and polymerization of methyl methacrylate ($C_5O_2H_8$) to poly methyl methacrylate $(C_5O_2H_8)_n$.
Initiation		initiated and activated by generating a free radical either chemically or with energy (such as heat, light, microwaves).
Propagation and termination		propagation via the binding of monomers followed by termination through shifting of the free electrons to the chain end

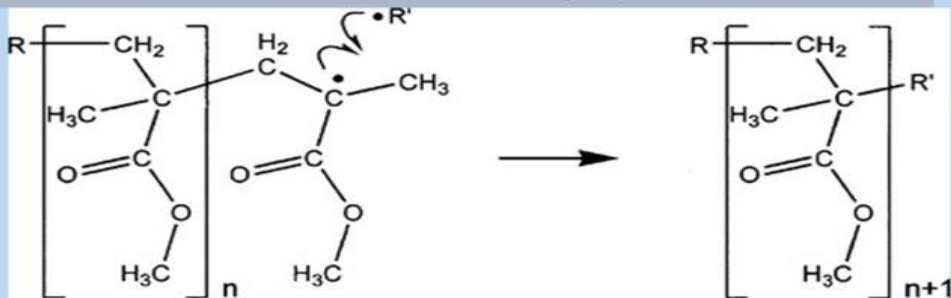
Inhibition

- 1- the addition of **chemical inhibitors (such as hydroquinone)** to the monomer.
- 2-Also, **any impurities** present in the monomer may inhibit the polymerization by reacting with the free radicals

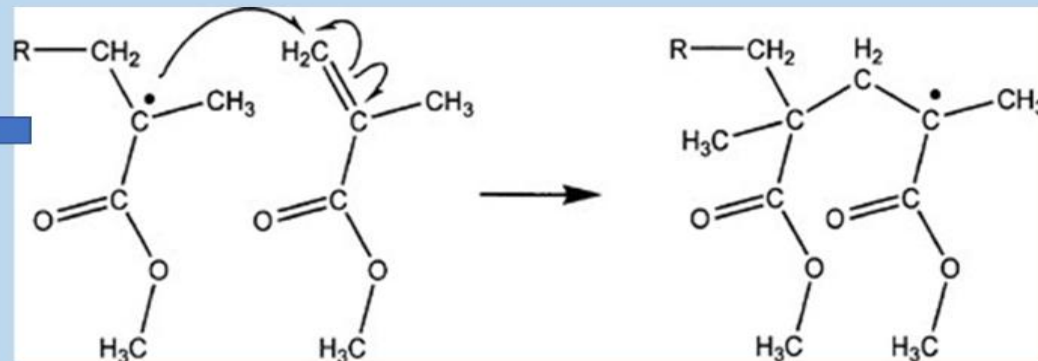


Initiation of polymerization of methyl methacrylate by the attack of a free radical on the double bond. (Single-barbed curly arrows indicate the shift of single electrons.)

The dissociation of benzoyl peroxide.



Chain termination results from the mutual reaction of two free radicals. Notice that R' can be any free radical, including a live polymer chain

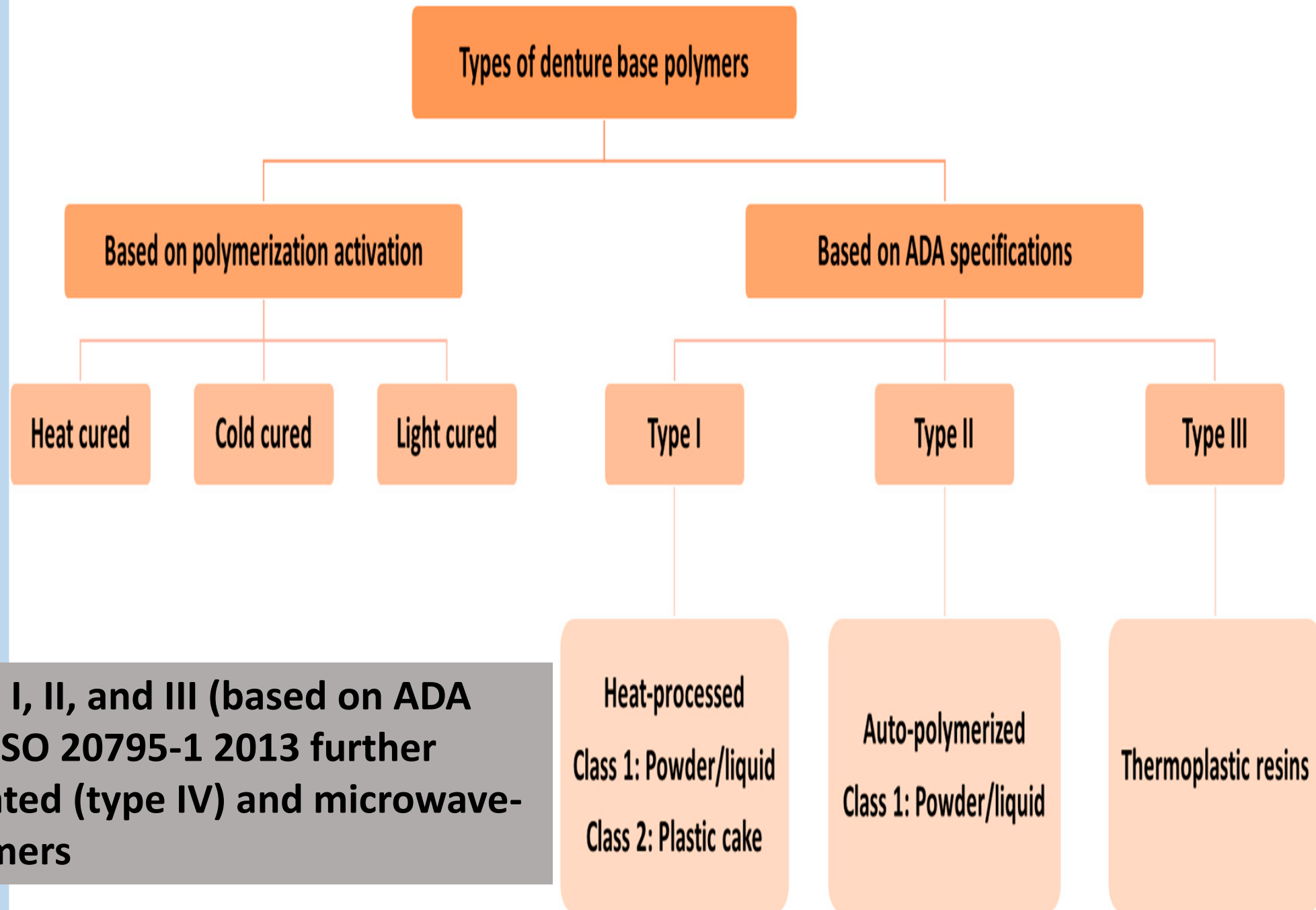


Chain propagation repeats the process

Classification:

Classification of denture base polymers based on polymerization activation and according to the ADA specifications

In addition to types I, II, and III (based on ADA specifications), the ISO 20795-1 2013 further included light-activated (type IV) and microwave-cured (type V) polymers



Heat cured PMMA

Commonly in the form of powder and liquid

PMMA powder: contains PMMA, benzoyl peroxide initiator, a plasticizer (dibutyl phthalate), opacifiers, fibers, and pigments or dyes

The liquid component: contains methyl methacrylate (MMA) monomer, ethylene glycol dimethacrylate as a cross-linking agent and hydroquinone as an inhibitor



<https://pocketdentistry.com/prosthetic-polymers-and-resins>
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polymerization in heat cured PMMA

- - The polymerization reaction starts upon mixing of both powder and liquid and requires heat energy (such as from a water bath) to activate the initiator.
- - benzoyl peroxide acts as an initiator, which dissociates into carbon dioxide (CO₂), generating free radicals upon heating

Various heat cycles used for the polymerization of heat-cured PMMA.

Heat Cycles	Temperature	Time	Terminal Boiling
1	74 °C	8 h	None
2	74 °C	8 h	1 h
3	74°C	3 h	1 h
4	73.9 °C	12 h	None
5	70 °C	3 h	1 h
6	100 °C	20 min	None

Cold-cured PMMA

- Also known as chemically cured or auto-polymerizing or self cured PMMA
- Does not require thermal energy
 - Initiator : A tertiary amine initiator such as dimethyl-p-toluidine activates the benzyl peroxide, chemically generating free radicals to initiate the polymerization

Applications: Due to its limitations, currently cold-cured PMMA materials are only used for: **fabricating custom trays - for denture repair**



Comparison between heat cured and self cured PMMA

Heat cured PMMA

- Higher degree of conversion
- Better mechanical properties
- Benzoyl peroxide is the initiator activated by heat
- Better color stability
- Lesser leaching of residual monomer
- Used as definitive denture base material

Self cured PMMA

- 1- Better dimensional stability and adaptation,
- 2- Minor polymerization shrinkage
 - lesser degree of conversion and polymerization
 - poor mechanical properties, such as the strength and hardness
 - the amine initiator oxidizes with time, resulting in discoloration , poor color stability
 - Higher residual monomer leaching

Light-Cured PMMA

- Light-cured (also known as visible-light-cured) PMMA is cured when exposed to visible light with certain wavelength.
- The PMMA is modified by replacing the conventional initiator with a *photo-sensitive agent (camphorquinone)*, which is activated and generates free radicals upon exposure to light.

Presentation: supplied in pre-mixed form containing PMMA fillers, silica, urethane dimethacrylate matrix, and acrylic resin monomers

Curing : To cure PMMA completely, the materials must be exposed to visible light for the required duration of time.

Advantages of light cured PMMA: easier fabrication

-polymerization shrinkage, the existence of residual monomers, and bacterial adhesion are lower when compared to heat- and cold-cured PMMA materials

Disadvantages of light cured PMMA: limited curing depth, technique sensitivity, and high cost .

The mechanical properties of light-cured PMMA materials are slightly inferior to conventional PMMA

Light cured PMMA applications are limited to relining , repair of dentures and fabrication of custom trays



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Copolymers

- Copolymers are fabricated to modify the physical and chemical properties of the polymer network.
- For example, in denture bases, copolymerization of the monomers of methyl methacrylate + butyl methacrylate results  in a copolymer that is more fracture resistant.
- Addition of comonomers such as octyl methacrylate makes the copolymer soft and flexible, and these copolymers are used for denture soft liners.
- Addition of hydroxyethyl methacrylate increases the wettability of the polymer by water and improves complete denture retention in a patient with xerostomia (dry mouth).



Ideal Properties of Polymer denture base materials

The PMMA-based materials should have certain desired properties depending on the biological application

- **Biocompatibility is the most important biological property,**

MUST BE:1- biocompatible and should not cause any irritation, toxicity, carcinogenicity or mutagenicity to the oral tissues .

- **2-insoluble in saliva and oral fluids.**
- **3-non-reactive to nutrients**

- **Residual monomer causes :mucosal irritation , tissue inflammation, and cytotoxicity HOW TO AVOID????**

1- Avoid adding a larger amount of monomer solution while mixing PMMA

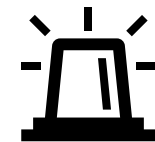
2- extended polymerization cycle may also decrease the residual monomer quantity and cytotoxicity compared to short curing cycles

3- The immersion in water prior to use may reduce the residual monomer amount, as well as the cytotoxicity of PMMA

4- heat-cured eluted lower amounts of monomers and has lower cytotoxicity compared to cold-cured PMMA

Occupational safety:

- The volatile nature of monomers and the dusty nature of the fine PMMA particles may result in professional hazards to the dental staff .
- -To reduce the concentration of monomer vapors and exposure in dental clinics and laboratories, appropriate safety measures are essential :- such as the use of personal protection (mask, gloves, goggles, face shield) and airtight containers, , good ventilation, and quick disposal of any spills .
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- -In addition, dental staff should be educated about the toxic and volatile nature of monomer solution (no exposure to flames) and immediate washing of the skin if exposed to the monomer solution + medical advice in case of direct exposure
- -Chronic inhalation of methyl methacrylate should be avoided because of potential toxicity to the liver



II-Ideal Mechanical properties of polymer denture base materials

Property	Description	Clinical relevance
Elastic modulus	Stiffness	High modulus is desirable to prevent flexure of denture or other prosthesis. Acrylic polymers have relatively low elastic moduli. Higher moduli are clinically desirable.
Impact strength	Resistance to fracture upon dropping	Important to resist fracture when dropping or blow to facial area. Acrylic polymers have moderate impact strength. Cross-linking increases by two-fold. Higher impact strengths are clinically desirable.
Fatigue strength	repeated stresses	Important to resist fracture under long-term intraoral forces (chewing, clenching). Acrylic polymers have moderate fatigue strength. Crosslinking can increases fatigue strength. Higher fatigue strengths are clinically desirable .

Acrylic polymers used for complete and partial dentures are relatively low in strength and modulus, brittle, and soft (DISADVANTAGE)

- The addition of butadiene-styrene rubber fillers to PMMA improves the impact strength significantly. Because these fillers also decrease the modulus of the base, rubber-modified denture bases are only used in patients who have a high risk of dropping the denture or experiencing facial blows or excessive occlusal forces

A high value of *modulus of elasticity* is advantageous.

A high value of *elastic limit* is required to ensure that stresses during biting and mastication do not cause permanent deformation.

A combination of a high modulus and high value of elastic limit would allow the base to be fabricated in thin section.

Fractures of upper dentures occur through the midline of the denture, due to flexing .

The denture base should have sufficient *flexural strength* to resist fracture.

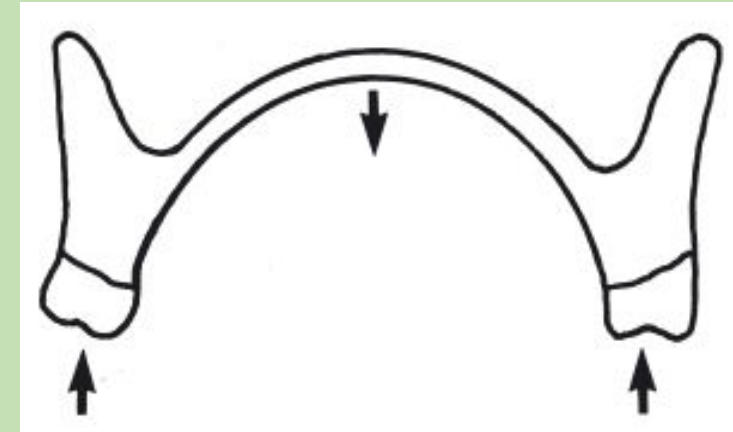
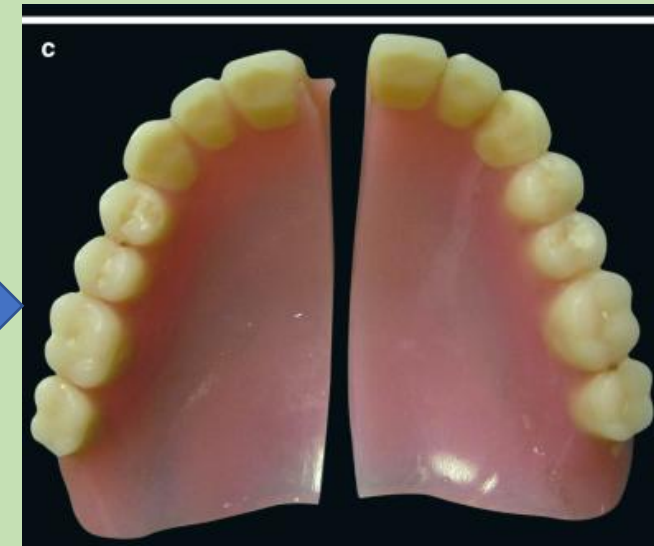
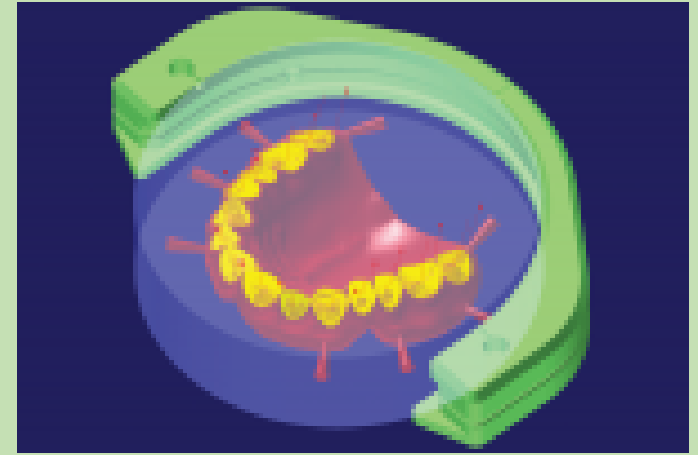


Diagram illustrating how an upper denture may flex about the midpoint of the palate. This fatigue process may eventually cause fracture



CAD/CAM PMMA

- More recently, several researchers investigated the use of CAD/CAM technologies for the fabrication of PMMA dental prostheses.
- In contrast to the conventional “flask–pack–press” technique, CAD/CAM techniques commonly use rapid prototyping and milling techniques.
- Although the chemistry of CAD/CAM PMMA is similar to that of conventional heat-cured PMMA, some authors claim that CAD/CAM PMMA shows superiority in terms of many properties, including its hardness, flexural strength, flexural modulus, and impact strength.



<https://www.aegisdentalnetwork.com/idt/2018/04/a-simplified-approach-to-digital-dentures>

Soft liners

Denture soft liners are materials that coat the tissue-bearing surface of a complete or partial denture.

these materials are soft or rubbery and are less likely to traumatize the oral soft tissues.

Soft liners are used for patients with severe undercuts of the edentulous ridges or for those patients whose ridges are continually sore.

They also are useful as a tissue treatment after oral surgery and in obturators for congenital or acquired defects of the palate, such as cleft palate or post surgery for oral-facial cancer.

SOFT LINERS

**LONG TERM
SOFT LINERS
(months)**

**SHORT TERM
SOFT LINERS
(days)**

**Methacrylate-
based
materials**

**Silicone-
based
polymers**



<https://www.dentistrytoday.com/application-of-a-long-term-resilient-lining-material/>**PROSTHODONTICS**Application of a Long-Term Resilient Lining Material

I- LONG TERM SOFT LINERS:

- Long-term materials are commonly methacrylate- or silicone-based polymers.
- A-Methacrylate-based materials: often consist of poly(methyl/ethyl methacrylate) copolymers with plasticizers.
- Plasticizers are not generally chemically bonded to the polymer network
- Methacrylate-based liners are cured either by heat or chemically at room temperature, depending on their formulation.
- Heat-cured formulations are processed in a laboratory, whereas chemical-cure materials may be processed either in the laboratory or at chairside.

The hardness of the long-term acrylic materials increases slowly as the plasticizer is leached out by exposure to oral fluids.

B-Silicone-based polymers :

They are also used as long-term soft liners.

Silicone soft liners are either heat cured or chemical cured, but they are processed in a laboratory regardless of the initiation mechanism.

Unlike methacrylate-based liners, silicone liners do not harden as time passes because they contain no plasticizer.

Clinically, silicone liners may support the growth of Candida albicans (yeast)



<https://doi.org/10.5272/jimab.2017234.1813>

Journal of IMAB - Annual Proceeding (Scientific Papers), 2017 Oct-Dec;23(4)

Original article

PRESENCE OF *CANDIDA* SPP. IN THE SALIVA OF PATIENTS WITH COMPLETE DENTURES, LINED WITH SILICONE-BASED ELASTIC MATERIALS.

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II- SHORT TERM SOFT LINERS:

Short-term materials are used in clinical situations in which the **edentulous** tissues are inflamed or swollen

Short-term lining materials consist of a polyethylene methacrylate powder and liquid plasticizers. When these materials are mixed, there is no polymerization reaction

Short-term materials, also known as “treatment” materials, are always applied at chairside, and the liner is replaced frequently (every 3 days or sooner)

They become harder as the oral fluids leach out the plasticizers.



Denture care

Complete or partial dentures should be removed each night and cleaned to ensure the health of the supporting oral tissues.



Q:When removed from the mouth, dentures must remain wet and are best stored in water. Why??

Answer: 1-Because acrylic polymers have high water sorption, desiccation results in a significant loss of water and shrinkage of the polymer.

2-If cycles of hydration and desiccation continue, the denture will gradually distort and crazing would appear.

HOW TO CARE????????????

-Acrylic polymers should be cleaned with a soft brush and mild abrasives, more aggressive cleaning will scratch the soft polymer .



-Disinfection of the acrylic polymer For complete dentures without a metal component, overnight immersion in diluted sodium hypochlorite will disinfect a denture.



-However, the use of hypochlorite is contraindicated in appliances that contain alloys as part of the prosthesis, because hypochlorite will corrode the alloy.



Patient must rinse the denture from cleaning or disinfectant solution by running water before putting it in his/her mouth



WHAT TO AVOID DURING CARE????????????

-Dentures should not be exposed to alcohol or acetone, as these or other solvents will dissolve and swell the polymer network and damage the denture surface.



-Denture acrylic polymers should never be cleaned or placed in hot (greater than 35° C [95° F]) water.

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• **THANK YOU**